From Perceptrons to Pre-Trained Transformers

A Mathematical & Systems Introduction to Machine Learning, Deep Learning, and LLMs

Autonomous Systems & Neural Foundations Engineering

October 14, 2026

Abstract

This monograph provides an end-to-end, mathematically rigorous yet visually intuitive introduction to the hierarchy of modern artificial intelligence: Classical Machine Learning (ML), Deep Neural Representations (DL), and Large Language Models (LLMs). We dissect learning paradigms, optimization algorithms, gradient-based dynamics, statistical evaluation metrics (Confusion Matrix, Precision/Recall, F_1, Z-scores, K-Fold cross-validation), the Transformer self-attention mechanism, Grouped-Query Attention (GQA), and decentralized pre-training from zero weights. Complete with self-contained vector TikZ diagrams, this reference is designed for computer scientists, systems programmers, and machine learning practitioners.

Contents

1 The AI Hierarchy: Foundations & Taxonomy **1**
1.1 Core Learning Paradigms 2

2 Training Dynamics: Epochs, Batches, and Iterations **2**

3 Diagnostic Metrics, Statistical Scores & Validation **3**
3.1 The Visual Confusion Matrix 3
3.2 Precision, Recall, and the F_1 Score 4
3.3 K-Fold Cross-Validation 4
3.4 Standardization via Z-Scores 5

4 Deep Representations: The Transformer Engine **5**
4.1 Attention Evolution: MHA vs. MQA vs. GQA 5

5 The Complete Foundation Model Lifecycle **5**
5.1 Inference Decoding Strategies 5

6 Comprehensive Diagnostic Formulary **7**

---

1 The AI Hierarchy: Foundations & Taxonomy

Artificial Intelligence is organized as concentric paradigms, where each domain builds directly on the mathematical primitives of its predecessor.

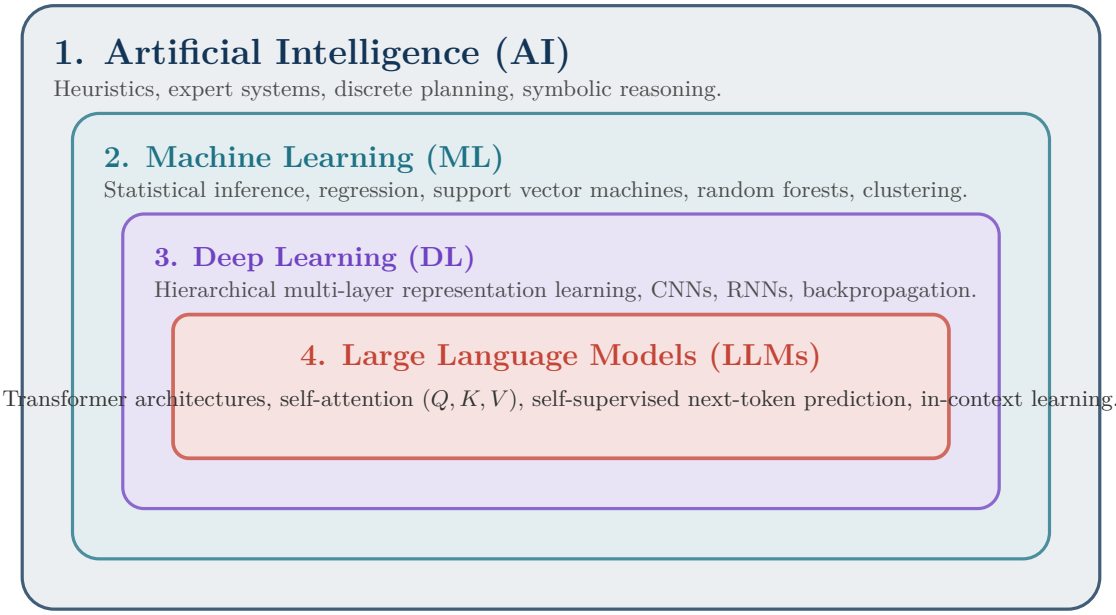


Figure 1: The Architectural Hierarchy: From Symbolic AI to Foundation LLMs.

1.1 Core Learning Paradigms

Machine learning algorithms are classified into three primary operational paradigms:

1. **Supervised Learning**: Given a dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ of features $\boldsymbol{x}_i \in \mathbb{R}^d$ and ground-truth targets y_i, find a parameterized mapping $f_{\boldsymbol{\theta}} : \mathbb{R}^d \to \mathcal{Y}$ that minimizes empirical risk:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i), y_i) \tag{1}$$

2. **Unsupervised Learning**: Given unlabeled observations $\mathcal{D} = \{\boldsymbol{x}_i\}_{i=1}^N$, infer latent structures, cluster assignments, or low-dimensional manifolds without ground-truth labels $(p(\boldsymbol{x}))$.

3. **Self-Supervised Learning (Foundation of LLMs)**: Targets are generated automatically from the structure of the data itself. For language modeling, given a token stream $(w_1, w_2, \ldots, w_T)$, the task is autoregressive next-token prediction:

$$\max_{\boldsymbol{\theta}} \sum_{t=1}^{T} \log P(w_t \mid w_1, w_2, \ldots, w_{t-1}; \boldsymbol{\theta}) \tag{2}$$

2 Training Dynamics: Epochs, Batches, and Iterations

A common source of confusion in optimization is the relationship between an *Epoch*, a *Batch*, and an *Iteration (Step)*.

Definitions at a Glance

- **Batch Size** (B): The number of distinct input sequences processed in parallel across compute cores before performing a gradient update.
- **Iteration / Step** (S): Exactly **one parameter update** via gradient backpropagation:
$$\boldsymbol{\theta}_{t+1} \leftarrow \boldsymbol{\theta}_t - \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_B(\boldsymbol{\theta}_t)$$
- **Epoch** (E): Exactly **one complete traversal** over every sample in the training corpus:
$$\text{Iterations per Epoch} = \left\lceil \frac{N_{\text{total}}}{B_{\text{effective}}} \right\rceil$$

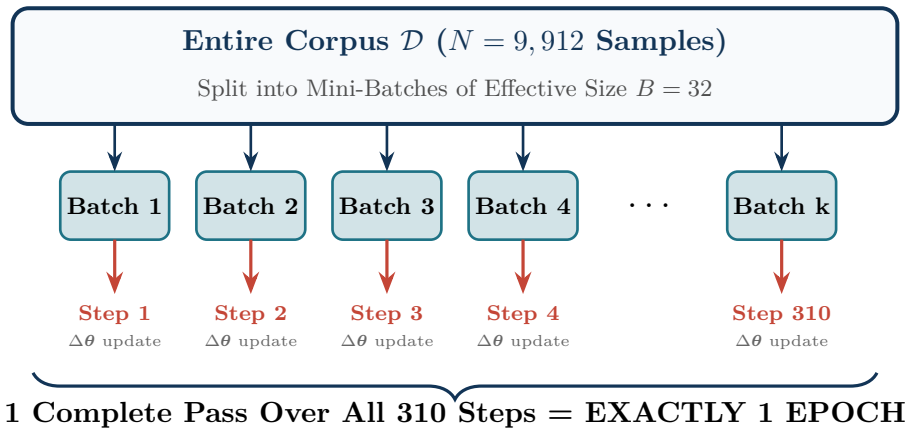


Figure 2: The Decomposition of an Epoch into Discrete Optimization Iterations.

3 Diagnostic Metrics, Statistical Scores & Validation

In systems diagnostics, intrusion detection, and classification, raw accuracy is frequently misleading due to severe class imbalance.

3.1 The Visual Confusion Matrix

Every binary decision boundary induces four distinct event classes. Figure 3 maps how Precision, Recall, and Accuracy are partitioned from this matrix.

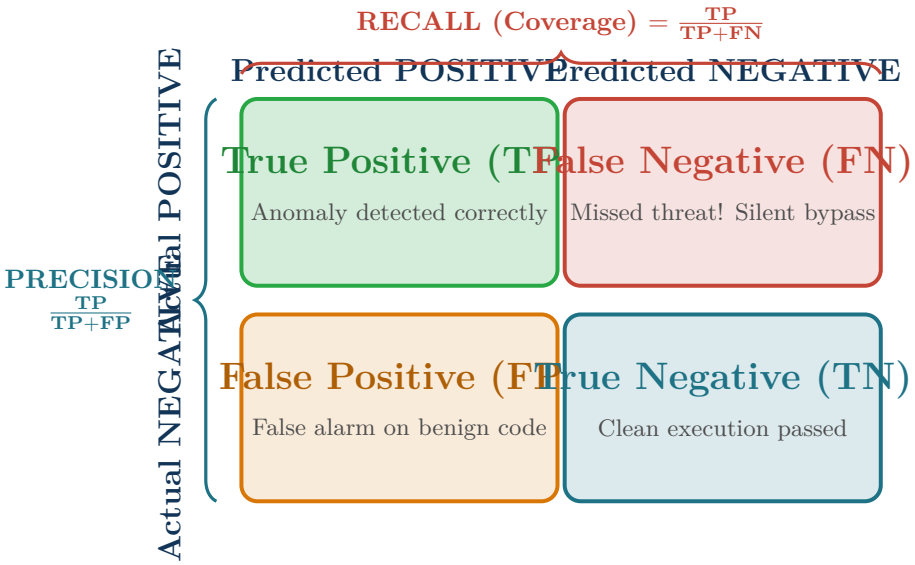


Figure 3: The Geometry of Precision and Recall on the Confusion Matrix.

3.2 Precision, Recall, and the F_1 Score

Fundamental Classification Formulas

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

$$\text{Precision (Positive Predictive Value)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (\textit{Alert Reliability}) \tag{4}$$

$$\text{Recall (Sensitivity / True Positive Rate)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (\textit{Threat Coverage}) \tag{5}$$

$$F_1 \text{ Score (Harmonic Mean)} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{6}$$

Why the Harmonic Mean? The arithmetic mean of $P = 1.0$ and $R = 0.0$ yields 0.5, masquerading as moderate performance. The harmonic mean forces $F_1 \to 0$ whenever either metric collapses:

$$F_\beta = (1 + \beta^2)\frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) + \text{Recall}} \tag{7}$$

When missing an event is catastrophic (e.g., zero-day exploit detection), we select $\beta = 2$ (F_2 score) to weight Recall twice as heavily as Precision.

3.3 K-Fold Cross-Validation

To avoid evaluation bias on arbitrary train-test splits, the dataset $\mathcal{D}$ is partitioned into K disjoint subsets $\mathcal{D}_1, \ldots, \mathcal{D}_K$. Figure 4 illustrates how every partition serves as an unseen test benchmark.

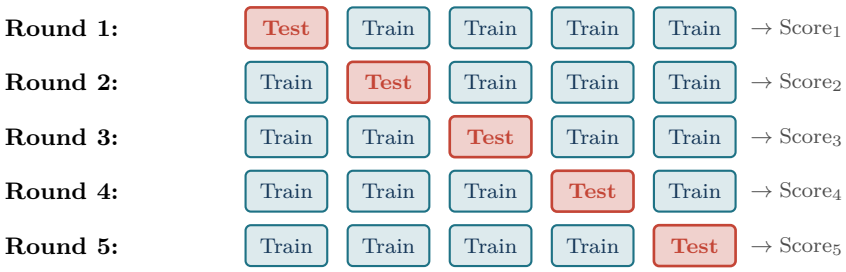


Figure 4: K-Fold Cross-Validation: Every sample is evaluated in held-out test data.

3.4 Standardization via Z-Scores

In anomaly detection, a Z-score measures how many standard deviations an observation x deviates from the baseline mean μ:

$$z = \frac{x - \mu}{\sigma} \tag{8}$$

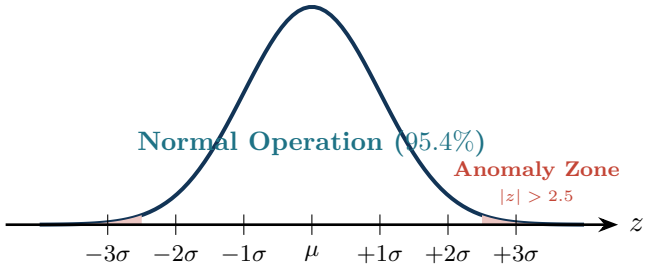


Figure 5: The Standard Normal Distribution and Statistical Z-Score Thresholds.

Under a normal distribution $\mathcal{N}(\mu, \sigma^2)$, values where $|z| > 3.0$ represent extreme tail events ($p < 0.0027$), flagging unauthorized runtime hooks, anomalous latency spikes, or structural shifts.

4 Deep Representations: The Transformer Engine

Modern foundation models abandon recurrence (RNN/LSTM) in favor of the fully parallelizable **Self-Attention** mechanism.

4.1 Attention Evolution: MHA vs. MQA vs. GQA

Standard Multi-Head Attention (MHA) maintains an independent Key and Value head for every Query head, causing massive Key-Value (KV) cache memory bloat during inference. Grouped-Query Attention (GQA) groups query heads to share a smaller number of KV heads:

5 The Complete Foundation Model Lifecycle

5.1 Inference Decoding Strategies

During autoregressive generation, logits $\boldsymbol{z}_t$ are converted to token probabilities via temperature scaling $T > 0$:

$$P(w_i) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \tag{9}$$

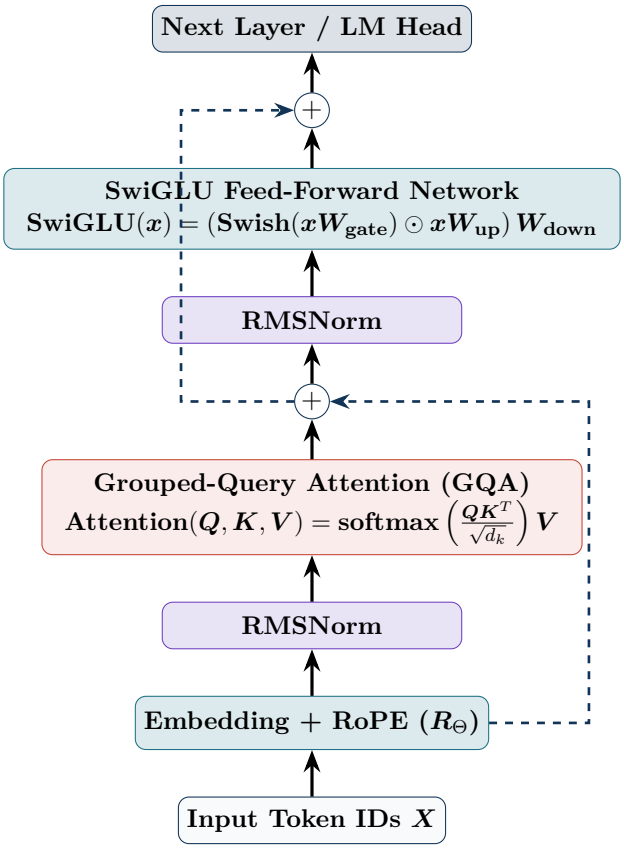


Figure 6: The Modern Pre-LayerNorm Transformer Block (LLaMA / DEMON Architecture).

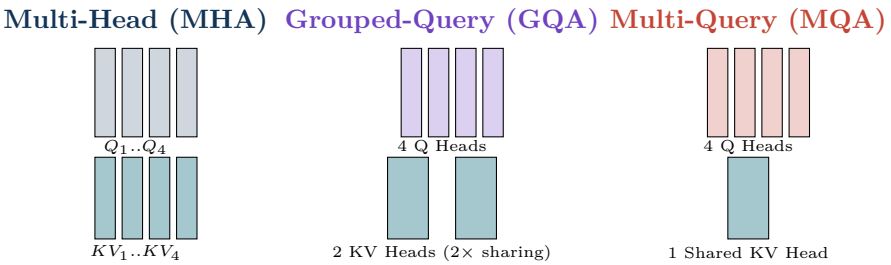


Figure 7: Comparison of Attention KV Cache Footprints (GQA balances speed and memory).

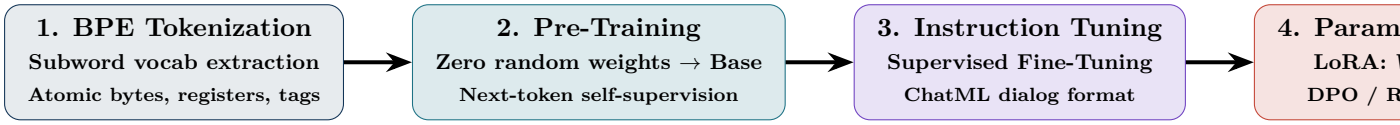


Figure 8: The End-to-End Foundation Model Lifecycle.

- **Greedy** ($T \to 0$): Always selects $\text{argmax}_i\, z_i$. Completely deterministic.
- **Top-k Sampling**: Truncates candidate pool to the k most probable tokens.
- **Top-p (Nucleus) Sampling**: Dynamically selects the smallest subset of tokens whose cumulative probability mass exceeds threshold p:

$$\sum_{i \in V^{(p)}} P(w_i) \geq p \tag{10}$$

6 Comprehensive Diagnostic Formulary

Concept	Mathematical Formulation	Systems / Diagnostic Interpretation
Precision	$\frac{\text{TP}}{\text{TP+FP}}$	Probability that an alert corresponds to an authentic anomaly.
Recall	$\frac{\text{TP}}{\text{TP+FN}}$	Probability that a true anomaly is successfully intercepted.
F_1 **Score**	$\frac{2\text{TP}}{2\text{TP+FP+FN}}$	Harmonic balance preventing over-optimization of false alarms or misses.
Z-Score	$z = \frac{x-\mu}{\sigma}$	Distance from execution latency mean in units of standard deviation.
Attention	$\text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V}$	Content-addressable soft associative memory across context tokens.
SwiGLU	$(\text{Swish}(\boldsymbol{x}\boldsymbol{W}_g) \odot \boldsymbol{x}\boldsymbol{W}_u)\boldsymbol{W}_d$	Bilinear gating mechanism amplifying salient token representations.
LoRA	$\boldsymbol{W} = \boldsymbol{W}_0 + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}$	Rank-r decomposition reducing trainable parameter footprint by $> 99\%$.

Table 1: Core Formulations in Machine Learning and Language Architectures.